COMP 3920

DATABASE SYSTEMS 1

PATRICK GUICHON

INDEXES

- Let's play a game.
- I'm thinking of a number
- Guess my number

1 to 100

INDEXES

- Answer:

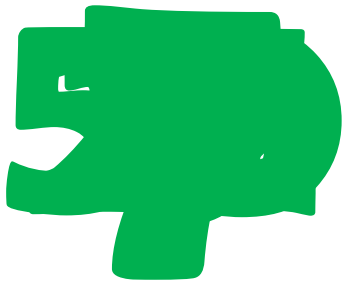
17

1 TO 100 GAME

- What's your first guess?

"50 is too small"

1-49



"50 is too big"
"25 is too big"
"12 is too big"
"6 is too small"
"9 is too big"
"8 is too big"
"7 is right!"

51-100

- What's your ***next*** guess?

1 TO 100 GAME

- If you do this right, you only need **7 guesses**.

- Why only 7 guesses?

  What is the $log_2 100$?

$\log_2 100$ = 6.64…
(at the most 7)

- Why not start by guessing 1, then 2, then 3… ?

1 TO 1,000,000 GAME

- What if I had a number between 1 and a million?
- Would you want to guess 1, then 2, then 3… ?

- What is the $log_2 1\ 000\ 000$?

$\log_2 1000000 = 19.93\ldots$
(at the most 20)

- If you do this right, you only need **20 guesses**!

ANIMAL TABLE

- Arctic Fox

- Common Name: Arctic Fox
- Genus Species: *Alopex lagopus*

INDEXES

Animal_ID	Common_Name	*Genus_Species*
1	Asian water buffalo	*Bubalus arnee*
2	Arctic fox	*Alopex lagopus*
3	Jaguarundi	*Felis yagouaroundi*
4	Bison, american	*Bison bison*
5	Mexican boa	*Boa constrictor mexicana*
6	Malachite kingfisher	*Corythornis cristata*
…	…	…
120	Verreaux's sifaka	*Propithecus verreauxi*

Page: 1 of 12

Rows: 1-120

INDEXES

Animal_ID	Common_Name	*Genus_Species*
121	Blue-faced booby	*Sula dactylatra*
122	Stork, jabiru	*Ephippiorhynchus mycteria*
123	Brazilian tapir	*Tapirus terrestris*
124	White rhinoceros	*Ceratotherium simum*
125	Snow goose	*Anser caerulescens*
126	Crake, african black	*Limnocorax flavirostra*
…	…	…
240	Goliath heron	*Ardea golieth*

Page: 2 of 12

Rows: 121-240

INDEXES

Can you find the *Genus Species* for the Ocelot?

Common Name: Ocelot
Genus Species: ??

Can you find the **Common Name** for the *Lama Pacos*?

Common Name: ??
Genus Species: *Lama Pacos*

http://www.thetimes.co.uk/imageserver/image/methode%2Ftimes%2Fprod%2Fweb%2Fbin%2F11fb3572-7380-11e7-8eac-856e9b33761e.jpg?crop=3000%2C1687%2C0%2C156

INDEXES

Table

Can you find the *Genus Species* for the Ocelot?

Common Name: Ocelot

Genus Species: ??

http://www.thetimes.co.uk/imageserver/image/methode%2Ftimes%2Fprod%2Fweb%2Fbin%2F11fb3572-7380-11e7-8eac-856e9b33761e.jpg?crop=3000%2C1687%2C0%2C156

INDEXES

Animal_ID	**Common_Name**	***Genus_Species***
1	Asian water buffalo	*Bubalus arnee*
2	Arctic fox	*Alopex lagopus*
3	Jaguarundi	*Felis yagouaroundi*
4	Bison, american	*Bison bison*
5	Mexican boa	*Boa constrictor mexicana*
6	Malachite kingfisher	*Corythornis cristata*
…	…	…
120	Verreaux's sifaka	*Propithecus verreauxi*

Page: 1 of 12

Rows: 1-120

INDEXES

Animal_ID	Common_Name	*Genus_Species*
121	Blue-faced booby	*Sula dactylatra*
122	Stork, jabiru	*Ephippiorhynchus mycteria*
123	Brazilian tapir	*Tapirus terrestris*
124	White rhinoceros	*Ceratotherium simum*
125	Snow goose	*Anser caerulescens*
126	Crake, african black	*Limnocorax flavirostra*
…	…	…
240	Goliath heron	*Ardea golieth*

Page: 2 of 12

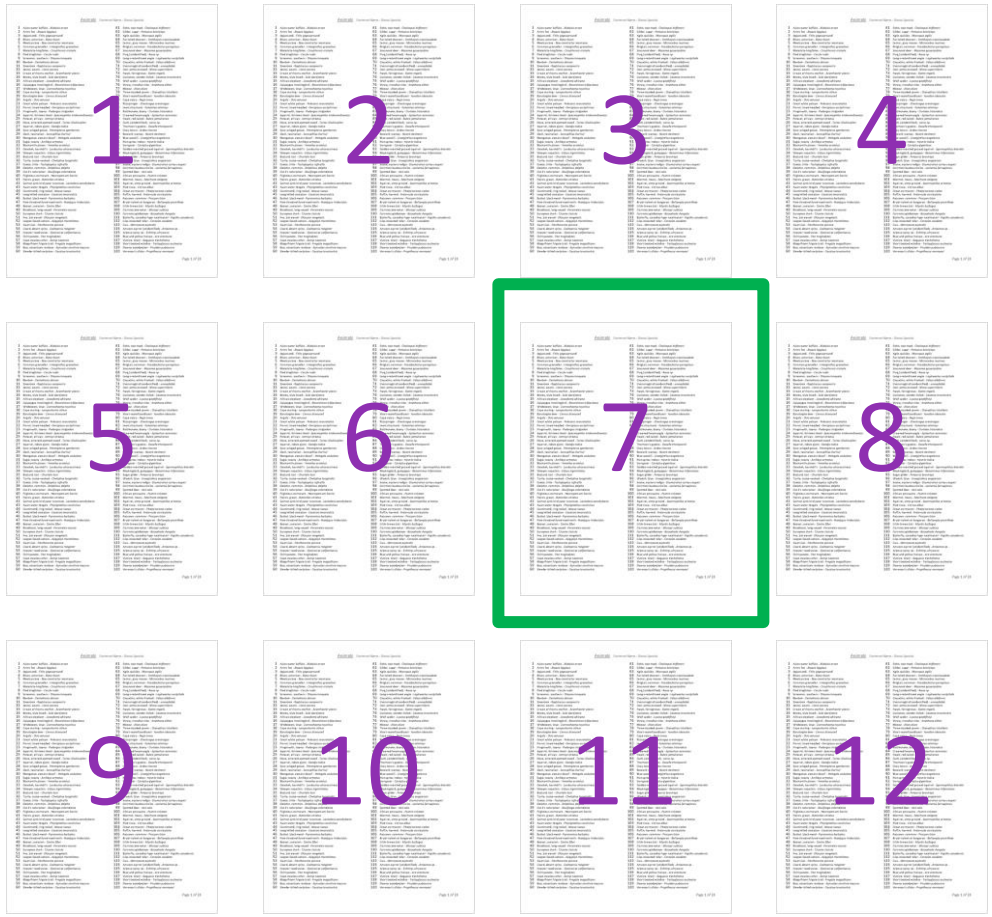
Rows: 121-240

INDEXES

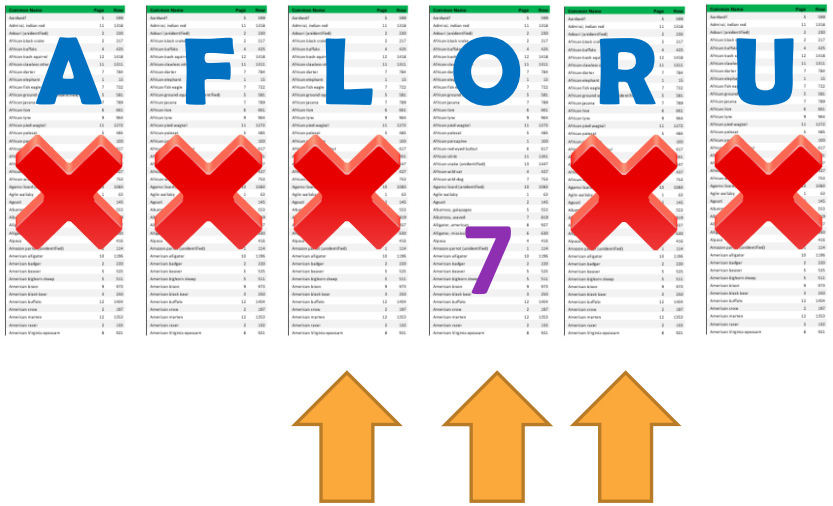
Common Name	Page	Row
Aardwolf	5	599
Admiral, indian red	11	1318
Adouri (unidentified)	2	230
African black crake	2	217
African buffalo	4	425
...	...	...
Zebra, common	2	188
Zebra, plains	12	1384
Zorilla	3	302
Zorro, azara's	4	368
Zorro, common	8	915

INDEXES

Table



Index



Sorted list (A-Z) of Common Name

INDEXES

Genus Species	Page	Row
Acanthaster planci	1	13
Acanthaster planci	13	1468
Acinynox jubatus	6	643
Acrantophis madagascariensis	3	259
Acrantophis madagascariensis	8	960
…	…	…
Zonotrichia capensis	5	499
Zosterops pallidus	12	1401
Zosterops pallidus	11	1269
Zosterops pallidus	12	1331
Zosterops pallidus	5	558

INDEXES

- Why doesn’t the Genus Species Index help us to find the Ocelot by Common Name?

- It’s not sorted by Common Name – We’d have to look at the whole thing to find the Ocelot.

INDEXES

- What is an index?
  - Helps optimize SELECT statements by speeding up

    WHERE clauses

    GROUP BYs

    Aggregate functions
- Just like a book index, they help find the data faster.

INDEXES

- SELECT genus_species
  FROM animal
  WHERE common_name = “Ocelot”

An index on Common Name Speeds up this query

An index on *Genus Species* does NOT help

INDEXES

- Why not index every column?
  - Each index must be maintained on each INSERT, UPDATE, DELETE on the table
    (slows down INSERTs, UPDATEs, DELETEs)
  - Indexes take space on storage media
    (sometimes as much as the table itself)
  - Indexes on the wrong column, don't help

INDEXES QUESTIONS

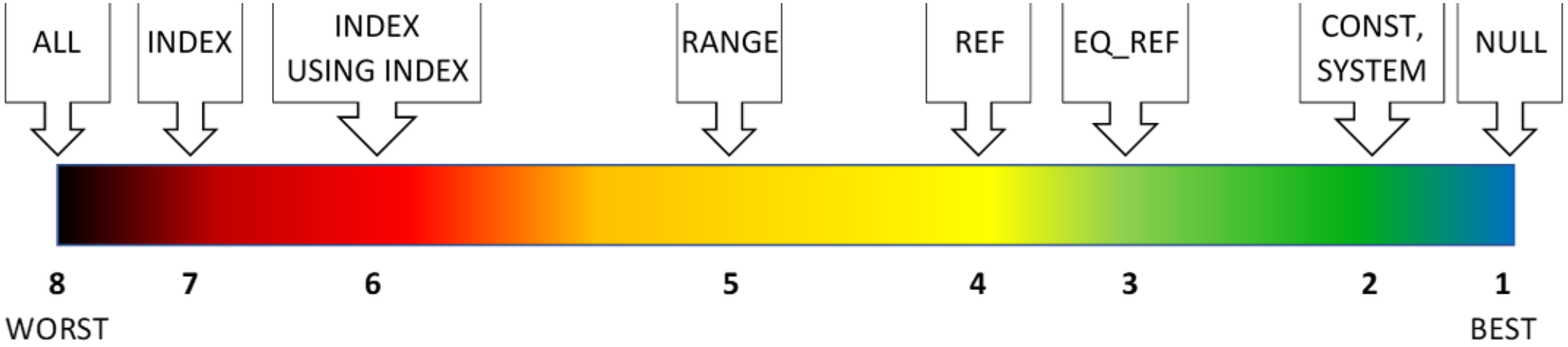
1. SELECT FROM largetable WHERE last_name=?
2. SELECT FROM largetable
   ORDER BY first_name, last_name
3. SELECT FROM largetable WHERE last_name=?
   ORDER BY first_name
4. SELECT FROM largetable
   WHERE birth_date >= '2000-01-01'
   ORDER BY last_name

INDEXES

- When do I need to index?
  - It can vary by table, but >= 10,000 rows
  - Lots of Joins
- How much space is used by an index?
- Should you index every column?

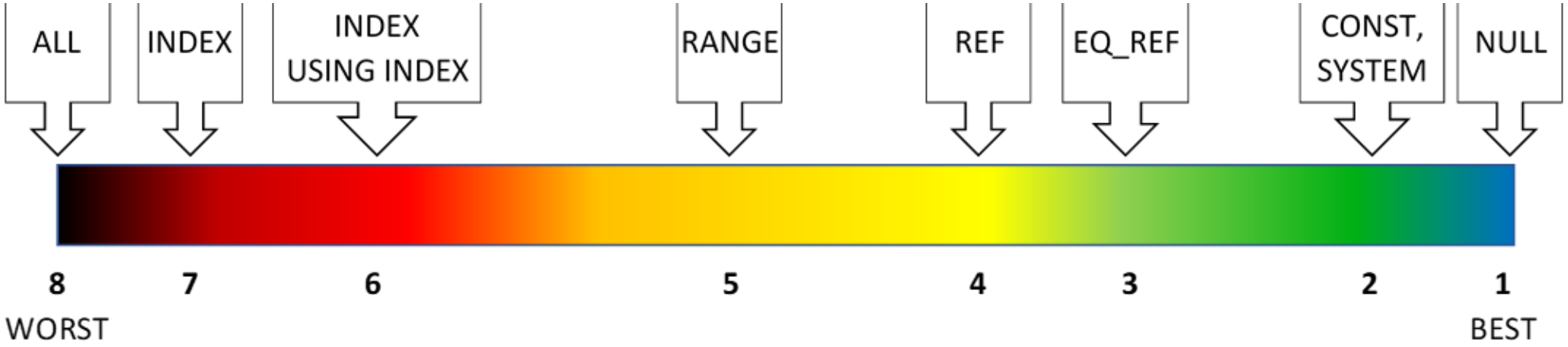
EXPLAIN COMMAND

```
+----+-----+------+---------------+------+-----+---------+-----+-------+
| id |     | type | possible_keys | key  |     | rows    |     | Extra |
+----+ ... +------+---------------+------+ ... +---------+ ... +-------+
|  1 |     | ALL  | NULL          | NULL |     | 1645301 |     | NULL  |
+----+-----+------+---------------+------+-----+---------+-----+-------+
```



EXPLAIN COMMAND

```
+----+-----+------+---------------+------+-----+---------+-----+-------+
| id |     | type | possible_keys | key  |     | rows    |     | Extra |
+----+ ... +------+---------------+------+ ... +---------+ ... +-------+
|  1 |     | ref  | name          | name |     |       1 |     | NULL  |
+----+-----+------+---------------+------+-----+---------+-----+-------+
```



EXPLAIN KEYWORD

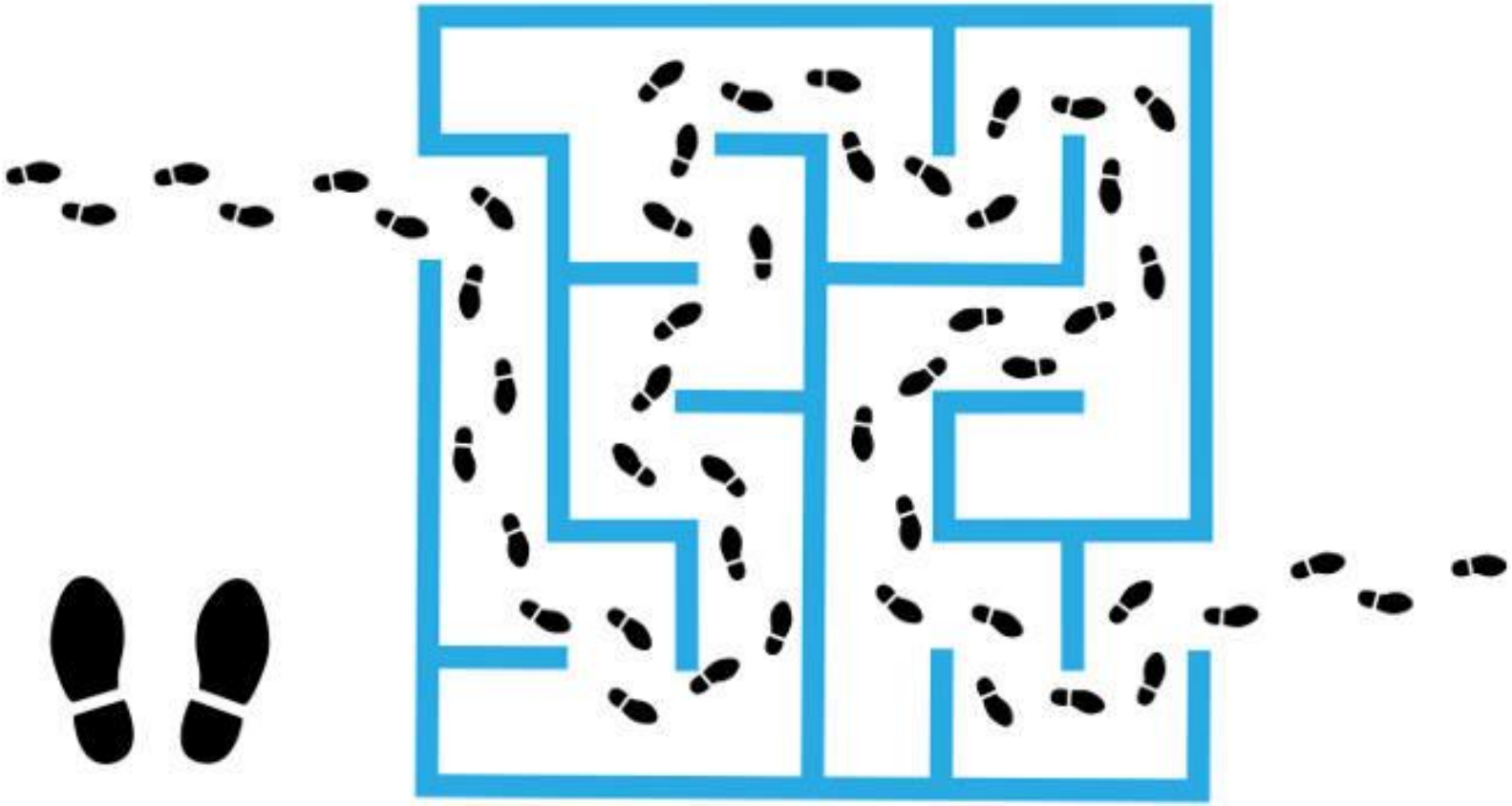
```
+----+-----+------+---------------+------+-----+---------+-----+-------+
| id |     | type | possible_keys | key  |     | rows    |     | Extra |
+----+ ... +------+---------------+------+ ... +---------+ ... +-------+
|  1 |     | ALL  | NULL          | NULL |     | 7243369 |     | NULL  |
+----+-----+------+---------------+------+-----+---------+-----+-------+
```

- Full Table Scan - Slow (lots of rows)

```
+----+-----+-------+---------------+------+-----+------+-----+-------+
| id |     | type  | possible_keys | key  |     | rows |     | Extra |
+----+ ... +-------+---------------+------+ ... +------+ ... +-------+
|  1 |     | ref   | name, age     | name |     |    2 |     | NULL  |
+----+-----+-------+---------------+------+-----+------+-----+-------+
```

- Indexed and Fast - (only a few rows)

LAB WALKTHROUGH



https://theeducatedlandlord.com/wp-content/uploads/2019/08/Tenant-Move-In-Landlord-tenant-walkthrough.jpg